

Alexander D. Gibson

BBiomedSc, MPhil, PhD Candidate

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1 Education

Doctor of Philosophy: Queensland University of Technology **2024 – Present**

- Thesis: “On Examining Poor Statistical Practices in Published Clinical Prediction Model Research”

Master of Philosophy: Queensland University of Technology **2022 – 2024**

- Thesis: “Standard Distance Triathlon: Examining Participation and Success, Predictive Performance Insights and Environmental Effects”

Bachelor of Biomedical Science: Queensland University of Technology **2019 – 2021**

- With distinction

2 Employment

PhD Candidate: Queensland University of Technology **Jul 2024 – Present**

- Successful recipient of a RTP scholarship valued at >\$126,632 for exceptional research potential.
- PhD project focused on improving poor statistical and research practices in prediction models.
- Studies conducted on fraudulent data, statistical hacking practices and empirical limits.
- Experienced understanding of Australian medical policy and Therapeutic Goods Administration legislation in the application of clinical prediction models and software and medical devices.

Research Assistant: Australian Centre for Health Services Innovation **Jan 2025 – Mar 2026**

- Project titled “Statistical Thinking In High-Performance Sport”.
- Working with the Chief Scientific Officer and Scientific Advisor of the Australian Institute of Sport to evaluate the quality of research produced by Australian Institute of Sport funded research.
- Collaborated on the development, pilot, collection of data and write up.

Clinical Research Assistant: Queensland University of Technology **Apr 2023 – Dec 2024**

- First round human clinical trial of micro-wearable biomedical technology.
- Management and conduct of clinical procedures, testing, safety, analyses and data management.
- Coordination and logistics management between sponsor personnel, internal colleagues and resources to meet timeline and budget targets.

Laboratory Assistant: Sullivan Nicolaides Pathology **Jun 2022 – May 2023**

- Leading local data collection for a partnered Franklin.ai \$130 million joint AI venture.
- Developed and implemented process to improve data throughput efficiencies by over 35%.

3 Funding

I have been successful in **\$167,356** of scholarships, grants and fundraising.

Australian Government Research Training Program Scholarship Jul 2024 – Present

- Successful recipient of a competitive RTP scholarship grant valued at more than **\$126,632** for exceptional research potential.

Association for Interdisciplinary Metaresearch and Open Science Oct 2025

- Successful applicant for **\$1,000** travel grant for a conference presentation.

National Breast Cancer Foundation Jul 2023 – Oct 2023

- Raised **\$37,107** running 12 marathons in 12 weeks.
- Channel 7 Sunrise live interview and featured article with QUT ([link](#)).

Starlight Children's Foundation Feb 2023

- Raised **\$2,117** with the SuperSwim event, swimming 65km the month of February.

The Fred Hollows Foundation Aug 2020 & Aug 2021

- Raised **\$500** with Fred's Big Challenge, running 141km the month of August.

4 Talks, Conferences & Podcasts

Bond University: Institute for Evidence-Based Healthcare Apr 2026

- Invited talk to present my PhD work and my first study on data provenance.

Metascience Matters Podcast Apr 2026

- Invited podcast discussing my research and expertise in clinical prediction models [link](#).

Association for Interdisciplinary Meta-Research and Open Science Nov 2025

- Selected lightning talk at the annual international conference.

QUT PhD Confirmation July 2025

- Presented and successfully completed my PhD research Confirmation of Candidature.

QUT MPhil Symposium Nov 2023

- Presented and successfully completed my Master of Philosophy research project.

5 Impact & Outreach

My research has led to updating of two guidelines for research and prediction models, I have consulted with government institutions in providing policy recommendations for protections of Australian research data as well as journal and publisher policy. My research has also led to the retraction of research, cleaning up the scientific literature.

Statistical Leadership Symposium	Jul 2026
• Organization and management of event logistics.	
Research Article Retractions	Mar 2026 – Present
• My first research article resulted in the retraction so far of three research articles. ■ Article 1, Article 2, Article 3	
Nature NEWS	Apr 2026
• My first research article was featured as a Nature story "Dozens of AI disease-prediction models were trained on dubious data" link. The article has an Altmetric score of 175.	
Retraction Watch	May 2026
• My first research article was featured as a Retraction Watch story "'Comically bad' datasets used to train clinical models for stroke and diabetes" link.	
Wikipedia: Kaggle	Apr 2026
• My first research article updated the Wikipedia page for Kaggle in a new section titled: "Medical Research Problems" link.	
Collection of Open Science Integrity Guides	Mar 2026
• My first research article updated the COSIG policy guidelines for scientific research integrity on formulaic research link.	
AusHSI HDR Working Group	Oct 2024 – Present
• Chaired and co-chaired meetings/workshop sessions for academic and professional development. • Organisation, management and development of professional development sessions. • Initiated the merger of two AusHSI working groups to improve efficiency and effectiveness across the management committee, internal working groups and HDR cohort.	
AusHSI Management Committee	Oct 2025 – Present
• Responsible for the representation of ~40 HDR students as well as ensuring the management, governance and strategic alignment AusHSI internally of QUT and in higher education. • Attending School of Public Health and Social work level academic forums showcasing my research and its impact.	
Personal Blog & Social Media	Oct 2024 – Present
• Yearly outreach of 70,000+ impressions focusing on science communication across LinkedIn and my personal blog.	
Queensland University of Technology	Apr 2025
• Identified outdated ethical information on QUT's Ethics Review Manager.	
World Medical Association	Apr 2025
• Identified an error in the Declaration of Helsinki. Alerted to the Senior Medical Advisor and Secretary General of the WMA which was subsequently and promptly corrected and updated.	
AIMOS Conference Committee Member	Nov 2023

- Managed and delivered components of the Association for Interdisciplinary Meta-Research & Open Science international conference.

World Triathlon

Mar 2024

- My research is showcased by the governing body, World Triathlon, and my research accounts for 66% of all showcased research on the World Triathlon website.

Queensland University of Technology

Sep 2023

- My fundraising running challenge was featured in a QUT article "Marathon man: Exercise science student running 12 marathons in 12 weeks" link.

6 Research Outputs

I have a total of 1 peer reviewed article and 4 current pre-registrations, a total of 7 citations and a H-Index of 1.



Preregistered



Open Data



Open Code



Preprint

Each research output has a badge representing if the study was preregistered, has available open access data, open access code and if the study was preprinted.

6.1 Publications



DN, Borg., **AD, Gibson.**, AJE, Bach., EM Beckman., SM, Tweedy., IB, Stewart., (2025). The Influence of Water and Air Temperature on Elite Wheelchair Triathlon Performance. *Temperature*, 11(4), 363–372. DOI: 10.1080/23328940.2024.2391170

K, Hutchins., BB, Sotomaior., **AD, Gibson.**, D, Borg., I Stewart (2024). Assessing The Response of Hyperthermic Men and Women to the Current Cold-Water Immersion Guidelines for Exertional Heat Stroke Treatment *Journal of Clinical Exercise Physiology*, DOI: 10.31189/2165-7629-13-s2.425

6.2 Preregistrations & Preprints



AD, Gibson., NM, White., GS, Collins., A, Barnett., (2025). Evidence of Unreliable Data and Poor Data Provenance in Clinical Prediction Model Research and Clinical Practice. *medRxiv*, DOI: 10.64898/2026.02.24.26347028



A, Barnett., **AD, Gibson.**, R, Tan., DN, Borg., P, Mensapa., (2025). Statistical Thinking in Australian High-Performance Sport. *OSF*



AD, Gibson., NW, White., DN, Borg., A, Barnett., (2026). Examining Hacking Practices In Published Clinical Prediction Models: A Protocol For A Cross-Sectional Observational Study. *OSF*, DOI: 10.17605/OSF.IO/B6RHJ



AD, Gibson., R, Blythe., NW, White., DN, Borg., A, Barnett., (2026). Examining Evidence for Empirical Limits in Clinical Prediction Models: A Cumulative Meta-Analysis. *OSF*.

6.3 Selected Blogs

Curiosity Chronicles

Mar 2026

- For The Patient or The Paper?: [link](#)

Australian Center for Health Services Innovation

Aug 2025

- A Model of Good Research Practice in Clinical Prediction: [link](#)

Curiosity Chronicles

Nov 2025

- Publishing in a Q1 Journal is not Q1 Research: [link](#)

7 Peer Review

I always complete an open-access peer-review where possible. I have completed a total of 3 rounds of peer review.

I have completed peer review for:

- The Journal of Diagnostic and Prognostic Research (open access peer-review)

8 Mentoring

PhD Student Mentor

Apr 2026

- Provided insights into the AusHSI HDR working group structure and advice on fostering a collaborative group of HDR students.
 - Bond University: Institute of Evidence-Based Healthcare PhD Students

Student Mentor

Feb 2026

- Providing mentorship support to students for career development.
 - Dimathi Liyanaarachchilage (Master of Public Health Student)

AusHSI Student Mentor

Jan 2026 – Present

- Providing mentorship support interally for new AusHSI HDR students.
 - Marla Donnino (Master of Science Student)

QUT HDR Supervisor Accreditation

Mar 2026

- Completed Research Integrity Training module.
- Completed Guide to HDR supervision at QUT module.

9 Professional Memberships

Statistical Society of Australia

2023 – Present

- Current student member of the Statistical Society of Australia

Association for Interdisciplinary Metaresearch and Open Science

2023 – Present

- Current member of the AIMOS association